For a general curve, however, the problem is more difficult, for example, how would you define the tangent lines shown in the following figures 10.2 to 10.4.

You might say that a line is tangent to a curve at a point P if it touches, but does not cross, the curve at point P. This definition would work for the first curve (Fig. 10.2), but not for the second (Fig. 10.3). Or you might say that a line is tangent to a curve if the line touches or intersects the curve exactly at one point. This definition would work for a circle but not for more general curves, as the third curve shows (Fig. 10.4).

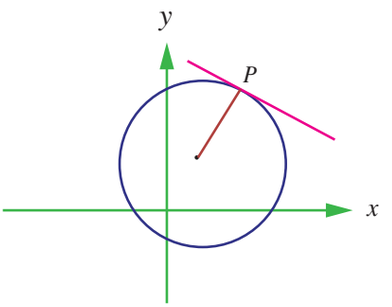


Fig. 10.1

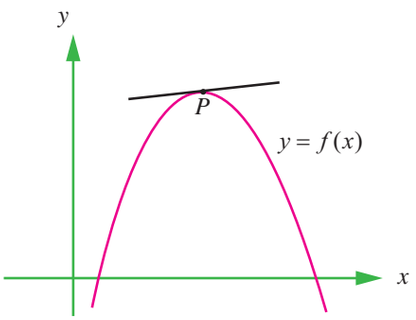


Fig. 10.2

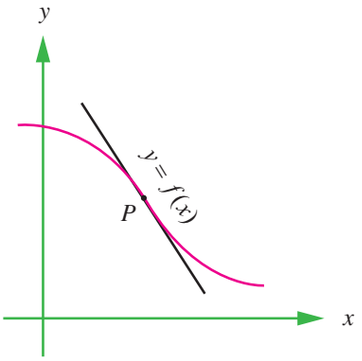


Fig. 10.3

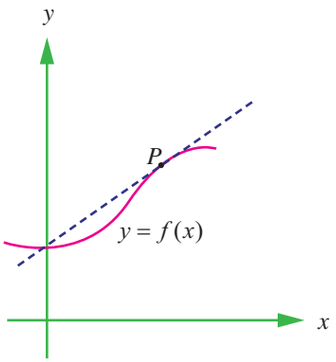


Fig. 10.4

Essentially, the problem of finding the tangent line at a point P boils down to the problem of finding the slope of the tangent line at point P. You can approximate this slope using a secant line through the point of tangency and a second point on the curve as in the following Fig. 10.5.

Let $P(x_0, f(x_0))$ be the point of tangency and $Q(x_0 + \Delta x, f(x_0 + \Delta x))$ be the second point.

The slope of the secant line through the two points is given by substitution into the slope formula

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m_{sec} = \frac{f(x_0 + \Delta x) - f(x_0)}{(x_0 + \Delta x) - x_0} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}.$$

That is, $m_{\text{sec}} = \dfrac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$, which is the slope of the secant line.

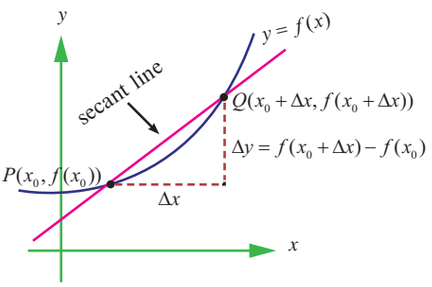


Fig. 10.5

The right hand side of this equation is a **difference quotient.** The denominator Δx is the **change in *x*** (increment in x), and the numerator $\Delta y = f(x_0 + \Delta x) - f(x_0)$ is the **change in *y***.

The beauty of this procedure is that you can obtain more and more accurate approximations of the slope of the tangent line by choosing points closer and closer to the point of tangency.

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